In addition, students must complete and defend an acceptable doctoral dissertation. Full-time PhD students must enroll in graduate colloquium (CBEN605) each semester. A maximum of 3 credits of CBEN605 can be counted toward the degree requirements.

Students in the PhD program are required to pass both a qualifying exam and the PhD proposal defense. After successful completion of 30 credits of coursework and completion of the PhD proposal defense, PhD candidates will be awarded a non-thesis master of science Degree. The additional requirements for the PhD program are described below.

PhD Qualifying Examination

The PhD qualifying examination will be offered twice each year, at the start and end of the spring semester. All students who have entered the PhD program must take the qualifying examination at the first possible opportunity. However, a student must be in good academic standing (above 3.0 GPA) to take the qualifying exam. A student may retake the examination once if he/she fails the first time; however, the examination must be retaken at the next regularly scheduled examination time. Failure of the PhD qualifying examination does not disqualify a student for the master's degree, although failure may affect the student's financial aid status.

The qualifying examination will cover the traditional areas of Chemical Engineering and will consist of two parts: GPA from core graduate classes (CBEN507, CBEN509, CBEN516, and CBEN518) and an oral examination. The oral examination will consist of a presentation by the student on a technical paper from chemical engineering literature. Students will choose a paper from a list determined by the faculty. Papers for the oral examination will be distributed well in advance of the oral portion of the exam so students have sufficient time to prepare their presentations. The student is required to relate the paper to the core chemical engineering classes and present a research plan, followed by questions from the faculty. A one- to two-page paper on the research plan is due the Friday prior to the oral examination.

If a student fails the first attempt at the qualifying exam, his/her grade from a 600-level Chemical Engineering elective can replace the lowest grade from the core graduate classes for, and only for, the GPA calculation defined above.

PhD Proposal Defense

After passing the qualifying exam, all PhD candidates are required to prepare a detailed written proposal on the subject of their PhD research topic. An oral examination consisting of a defense of the thesis proposal must be completed prior to their fifth semester. Written proposals must be submitted to the student's thesis committee no later than one week prior to the scheduled oral examination.

Two negative votes from the doctoral committee members are required for failure of the PhD proposal defense. In the case of failure, one re-examination will be allowed upon petition to the department head. Failure to complete the PhD proposal defense within the allotted time without an approved postponement will result in failure. Under extenuating circumstances, a student may postpone the exam with approval of the graduate affairs committee, based on the recommendation of the student's thesis committee. In such cases, a student must submit a written request for postponement that describes the circumstances and proposes a new date. Requests for postponement must be presented to the thesis committee no later than two weeks before the end of the semester in which the exam would normally have been taken.

Online graduate certificate in Chemical engineering processes in energy transitions

The Mines graduate certificate in Chemical Engineering Processes in Energy Transitions is a three-course online program that provides engaging learning experiences in understanding current challenges and existing technologies in the safe production of energy, development of efficient energy storage systems, and associated environmental remediation methods related to carbon capture and utilization. Courses in the program focus on real-world challenges and state-of-the-art technologies. By bringing salient aspects of energy and environment under one umbrella, students develop the expertise necessary to address technological problems, make economic decisions, or formulate government policies. This program is designed for professionals and recent graduates who want to acquire new skills for career advancement or get a head start on an advanced graduate degree. The certificate program requires three 3-credit graduate courses identified below.

CBEN522	CHEMICAL ENGINEERING FLOW ASSURANCE	3.0
CBEN598B	ADVANCED ELECTROCHEMICAL ENGINEERING	3.0
SYGN598A	NON-GEOLOGIC CCUS: CAPTURE & UTILIZATION	3.0

Courses

CBEN504. ADVANCED PROCESS ENGINEERING ECONOMICS. 3.0 Semester Hrs.
Advanced engineering economic principles applied to original and alternate investments. Analysis of chemical and petroleum processes relative to marketing and return on investments. Prerequisite: none. 3 hours lecture; 3 semester hours.

CBEN505. NUMERICAL METHODS IN CHEMICAL ENGINEERING. 3.0 Semester Hrs.
Engineering applications of numerical methods. Numerical integration, solution of algebraic equations, matrix 54 Colorado School of Mines Graduate Bulletin 2011 2012 algebra, ordinary differential equations, and special emphasis on partial differential equations. Emphasis on application of numerical methods to chemical engineering problems which cannot be solved by analytical methods. Prerequisite: none. 3 hours lecture; 3 semester hours.

CBEN506. ADVANCED FUNCTIONAL POROUS MATERIALS. 3.0 Semester Hrs.
Foundation on basic chemical strategies for making nanomaterials. Integration of fundamentals and functional applications of ordered porous materials at different length scales: from micro to macroporous regime. Chemical engineering concepts in nanochemistry. Existing and emerging functional applications of these porous materials in gas separations, heterogeneous catalysis, and adsorption.
Course Learning Outcomes

CBEN507. APPLIED MATHEMATICS IN CHEMICAL ENGINEERING. 3.0 Semester Hrs.
(I, II) This course stresses the application of mathematics to problems drawn from chemical and biological engineering fundamentals such as thermodynamics, transport phenomena, and kinetics. Formulation and solution of ordinary and partial differential equations arising in chemical engineering or related processes or operations are discussed. Prerequisite: Undergraduate differential equations course; undergraduate